Agus Fuad Mudhofar

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Education

•Bachelor of Computer Engineering *2022-Present*

Institut Teknologi Sepuluh Nopember, Surabaya GPA: 3.64

Personal Projects

•Camera Calibration and Image Processing Toolkit

A toolkit for camera calibration, image processing, and computer vision experiments using Python and React.

- Implements camera calibration using Python and OpenCV for accurate image measurements.
- Developed Harris Corner Detection and manual CNN experiments for image classification.
- Provides a React + TypeScript + Vite frontend for interactive visualization and demos.
- Technology Used: Python, React, TypeScript, OpenCV, NumPy, Vite.

•Smart Wheelchair with Eye Movement Control

A smart wheelchair system controlled by eye movement, integrating computer vision and embedded hardware.

- Developed a smart wheelchair system that allows users to control movement using eye gaze, enhancing accessibility for individuals with mobility impairments.
- Integrated real-time eye tracking and gesture recognition using Python and OpenCV, enabling intuitive navigation and control.
- Implemented embedded hardware components (Arduino/ESP32) for motor control and sensor feedback, ensuring reliable operation.
- Collaborated with a team to design, prototype, and test the system, resulting in a functional prototype showcased at the GEMASTIK XVII competition.
- Technology Used: Python, OpenCV, Arduino/ESP32, C/C++, computer vision algorithms.

•Sappy: Multi-Role Livestock Management App (Flutter)

A livestock management app for multiple roles (Farmer, Admin, Doctor) with NFC support.

- Provides role-based access and dashboards for farmers, administrators, and doctors, all in a unified mobile app.
- Supports English and Indonesian languages using built-in localization.
- Implements state management via Provider and enables NFC functionality for secure identification and tracking.
- Includes data visualization, authentication, and environmental configuration using modern Flutter ecosystem packages.
- Technology Used: Flutter, Dart, Provider, NFC, fl_chart, carousel_slider, custom fonts/assets.

•Dot Matrix Ping Pong Game (Embedded System)

A microcontroller-based two-player Ping Pong game using an 8x32 LED dot matrix and custom C++ firmware.

- Implements responsive paddle and ball mechanics, scorekeeping, and game flow on a chain of 8x8 LED matrix modules using the MD_Parola and MD_MAX72xx libraries.
- Features analog paddle control (potentiometers), digital smash/serve buttons, and real-time animated score display.
- Includes custom animations, win/lose logic, and a smash mechanic for increased gameplay excitement.
- Technology Used: C++ (Arduino/ESP32), MD_Parola, MD_MAX72xx, SPI, basic circuit design.

•OpenCV Poker Game

A Poker game with real-time card recognition using OpenCV and Python in a Jupyter Notebook.

- Implements real-time playing card detection and recognition using OpenCV image processing techniques, allowing the system to identify card values and suits from camera or image input.
- Integrates complete Poker game logic, including dealing, betting rounds, hand evaluation, and winner determination, all visualized and controlled through Jupyter Notebook cells.
- Facilitates interactive and step-by-step gameplay with visual feedback, supporting easy modification and educational exploration of both computer vision and game mechanics.
- Technology Used: Python, Jupyter Notebook, OpenCV, NumPy, image segmentation, template matching, Poker hand evaluation algorithms.

- **Alien Space (2D Game with C/C++ Graphics)**

  A classic-inspired 2D top-down space shooter game developed using C and C++ graphics programming.

  – Implements a real-time arcade-style game loop featuring player spaceship movement, alien enemies, bullet firing mechanics, and collision detection.
  – Renders game graphics using C/C++ graphics libraries, providing visual representation of player, aliens, bullets, and score.
  – Handles keyboard input for responsive control and integrates scoring and win/lose conditions.
  – Technology Used: C, C++, graphics.h (or SDL/SFML as applicable), structured programming, basic event handling.

- **Comic Reading Web Application**

  A web-based comic reading platform supporting manga, manhua, and manhwa with optimized reading experience.

  – Developed a responsive comic reading platform supporting manga, manhua, and manhwa with smooth page navigation and optimized image rendering.
  – Implemented core reading features including chapter management, reading history, authentication, and adaptive layouts for multiple devices.
  – Technology Used: React (frontend), Laravel (backend/API), PostgreSQL (database), with modular component architecture and optimized asset handling.

Experience

- **Research Intern Full Stack Developer** — *Feb July 2026*

  Bandung — Badan Riset dan Inovasi Nasional (BRIN) Pusat Riset Sains Data dan Informasi

  – Developed a crossplatform application (Mobile, Tablet, Desktop) using Flutter to precisely calculate NPK (Nitrogen, Phosphorus, Potassium) nutrient requirements for various vegetable crops.
  – Engineered a robust and scalable JSON REST API backend utilizing Laravel 12 and PostgreSQL to manage crop logic, calculations, and relational data.
  – Implemented secure, tokenbased user authentication and authorization workflows incorporating Laravel Sanctum to protect sensitive user data.
  – Designed an intuitive and highly responsive user interface in Flutter, ensuring a seamless and optimized experience across differing screen sizes and devices.
  – Architected the project codebase as a monorepo, efficiently streamlining development, testing, and deployment workflows for both the frontend and backend systems.

- **Research Assistant – Multimedia and IoT Lab (B300)** — *April - June 2025*

  Surabaya — Multimedia and IoT Laboratory (MIOT)

  – Conducted research on deploying deep learning models on edge devices using NVIDIA Jetson Nano, focusing on computer vision tasks.
  – Developed and optimized object detection pipelines using Python and C++ with MobileNet-SSD, YOLOv5n, and YOLOv11s architectures.
  – Explored model conversion and acceleration techniques with ONNX and NVIDIA TensorRT for real-time inference.
  – Integrated and evaluated models within the NVIDIA DeepStream SDK for efficient video analytics and deployment.
  – Benchmarked performance and resource utilization to optimize inference speed and accuracy in constrained environments.

- **Research Assistant – Smart Wheelchair Development** — *Aug 2024 - Present*

  Surabaya — Multimedia and IoT Laboratory (MIOT)

  – Maintained and enhanced a smart wheelchair system utilizing deep learning-based computer vision for hands-free navigation and control.
  – Integrated real-time eye-gaze and facial gesture recognition using Python and OpenCV, improving system responsiveness and user accessibility.
  – Collaborated with faculty and student teams to refine model architectures, optimize inference speed, and troubleshoot hardware-software integration issues.
  – Developed custom microcontroller firmware (C/C++) for reliable motor actuation and sensor feedback, ensuring safe and accurate wheelchair movement.
  – Documented system upgrades, conducted user testing, and presented results in lab meetings and academic showcases.

- **Lab Teaching Assistant – Basic Programming** — *Aug 2025 - Nov 2025*

  Surabaya — Multimedia and IoT Laboratory (MIOT)

  – Developed and maintained lab materials, including coding exercises and troubleshooting guides during practicum, while also being responsible for deploying and maintaining the DMOJ online judge website used in coursework.

- Guided students through practical assignments in C/C++, providing real-time support and technical explanations.
- Evaluated student submissions, provided constructive feedback, and contributed to grading rubrics to ensure fair and consistent assessment.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key programming concepts.

- **Lab Teaching Assistant – Digital Systems** *Mar 2025 - May 2025*
  Surabaya Robotics and Digital Systems Laboratory (B401)
  - Conducted lab sessions for *Rangkaian Digital*, supporting over 100 undergraduate students each semester.
  - Developed and maintained lab materials, including hardware schematics and troubleshooting guides for digital logic experiments.
  - Guided students through practical assignments in digital circuit design, providing real-time support and technical explanations.
  - Collaborated with faculty to improve lab workflows, implement new teaching tools, and enhance the overall learning experience.
  - Organized and led review sessions before exams, addressing common misconceptions and reinforcing key concepts in digital systems.

•Lab Teaching Assistant – Telemathics Workshop *Mar 2025 - May 2025*

Surabaya Robotics and Digital Systems Laboratory (B401)

- Prepared and developed practicum modules and instructional materials for *Workshop Telematika*, ensuring structured, hands-on learning for undergraduate students.
- Supervised and guided students during laboratory sessions, particularly in hardware-oriented activities including manual soldering, 3D design for prototyping, hot air rework (blower) soldering, reflow soldering, and 3D printing.
- Provided real-time technical assistance in C/C++ programming and microcontroller-based experiments, helping students troubleshoot both software and hardware issues.
- Assessed student practicum results, delivered constructive feedback, and contributed to grading criteria to ensure fair and consistent evaluation.
- Organized and led review sessions prior to assessments, reinforcing key concepts and practical skills.

•PIC IoT Division – TIM Kawal GEMASTIK *Jan 2025 - Nov 2025*

Surabaya Institut Teknologi Sepuluh Nopember (ITS)

- Served as the Person-in-Charge (PIC) for the Internet of Things (IoT) division, coordinating communication between faculty supervisors, professors, mentors, and participating student teams.
- Assisted in the selection and evaluation of candidate teams through proposal reviews, technical assessments, and readiness evaluations for the GEMASTIK competition.
- Provided administrative support for finalist teams, including documentation management, submission compliance, and coordination with the national GEMASTIK committee.
- Acted as the primary coordinator for mentoring activities by scheduling, organizing, and facilitating mentoring sessions with professors and domain experts in the IoT field.
- Supported teams' non-technical needs such as presentation design, report structuring, and competition readiness to ensure high-quality deliverables.

•Web Developer – MIOT Webpage *Aug 2024- Present*

Surabaya Multimedia and IoT Laboratory (MIOT)

- Designed and developed the official MIOT laboratory webpage to showcase research, publications, and team members.
- Implemented a responsive and modern user interface using React.js and Tailwind CSS, ensuring accessibility across devices.
- Integrated dynamic content management for news, events, and member profiles using JSON-based data structures.
- Collaborated with lab members to gather requirements, iterate on design, and deploy the site using GitHub Pages.
- Optimized site performance and SEO, and provided documentation for future maintenance and updates.

•Workshop Assistant – International Workshop on Deep Learning *24 - 29 Nov 2025*

Surabaya Institut Teknologi Sepuluh Nopember

- Assisted in the execution of a 5-day international workshop focused on machine learning and deep learning applications for assistive technology.
- Provided hands-on support to participants by helping resolve technical issues and learning difficulties during workshop sessions.
- Supported professors and speakers in running the event, including administrative coordination and on-site logistical tasks.
- Contributed to creating an inclusive and supportive learning environment for students from Malaysia, Vietnam, the Philippines, Ethiopia, Pakistan, and Indonesia.

•Representative – Toyota Exhibition Smart Wheelchair Demo *26 - 27 May 2025*

Jakarta Toyota Motor Manufacturing Indonesia

- Represented the smart wheelchair project at the Toyota Exhibition, showcasing the university's innovation in assistive technology.
- Assisted with technical preparation, including system integration, hardware checks, and software updates to ensure reliable operation during the event.
- Supported on-site troubleshooting, addressing real-time technical issues and adapting to exhibition constraints to maintain smooth demonstrations.
- Engaged with visitors, industry professionals, and Toyota representatives, providing detailed explanations of the wheelchair's computer vision-based control system and its societal impact.
- Helped prepare supporting materials such as technical documentation, user guides, and demonstration scripts to facilitate effective communication and audience understanding.

Positions of Responsibility

•Team Leader – GEMASTIK XVII (IoT Division) *Jan - Nov 2024*
Semarang Semarang State University (UNNES)

- Led a team representing Institut Teknologi Sepuluh Nopember in the Smart Devices, Embedded Systems, and IoT Division at GEMASTIK XVII.
- Coordinated the team's efforts across software and hardware components, managed timelines, and delegated responsibilities.
- Developed a smart wheelchair system controllable via eye movement, integrating computer vision and embedded hardware.
- Secured the National Finalist award (Juara Harapan) through efficient planning, ideation, and execution of an IoT-based innovation.

•Head of Logistics Department – Multimedia and Game Event (MAGE X) *Jan - November 2024*
Surabaya Multimedia and Game Event (MAGE X)

- Led the logistics team for MAGE X, a major university-level multimedia and game competition event.
- Coordinated procurement, inventory management, and distribution of event materials and equipment.
- Developed and executed logistics plans to ensure timely setup and smooth operation across multiple event venues.
- Collaborated with other departments to address real-time needs and resolve logistical challenges during the event.
- Managed vendor relations, budget tracking, and post-event inventory reconciliation.

•Expert Staff - UKM Rebana ITS *Aug - Sep 2024*
Surabaya Student Activity Unit Rebana ITS

- Served as a mentor guiding project leaders and team members in executing their respective work programs.
- Oversaw planning, execution, and evaluation phases to ensure alignment with organizational goals.
- Facilitated knowledge sharing and skill development through workshops and training sessions.

•Event Coordinator - Welcome Party *Jul - Aug 2023*
Surabaya Studet Activity Unit Rebana ITS

- Led the organizing committee to successfully hold the Welcome Party for new members.
- Managed team operations, chaired meetings, and delegated responsibilities to ensure event objectives were met.
- Coordinated logistics, marketing, and participant engagement to create a welcoming and inclusive atmosphere.

Technical Skills and Interests

Languages: C/C++, Python, Javascript, HTML+CSS
Libraries : C++ STL, Python Libraries, ReactJs
Web Dev Tools: Nodejs, VScode, Git, Github
Frameworks: ReactJs
Cloud/Databases:MongoDb, Firebase, Relational Database(mySql)
Relevent Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Software Engineering.
Areas of Interest: Sofware Development
Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability